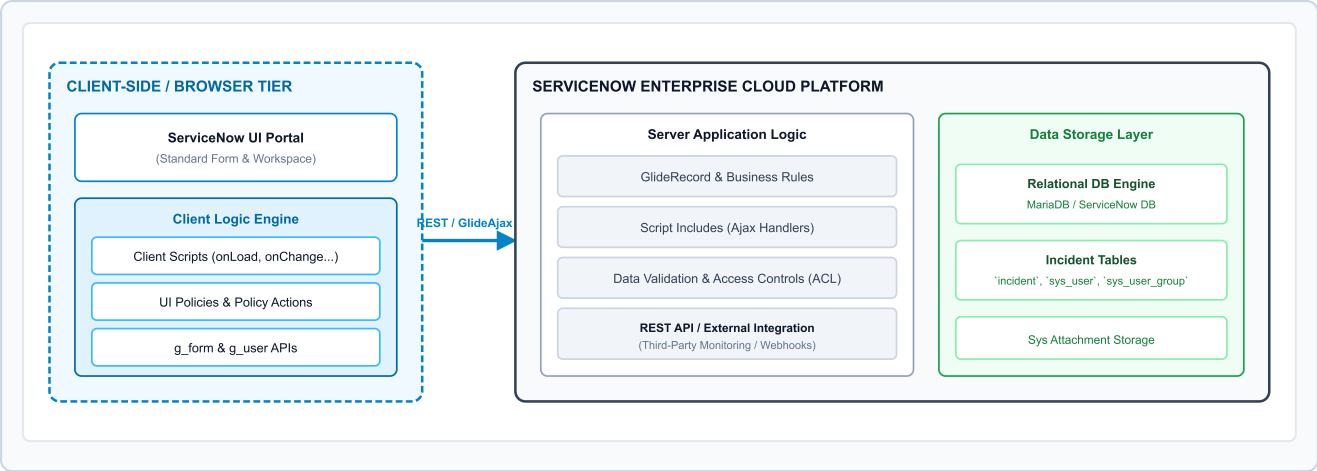


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date:		Maximum Marks:	
Team ID:	SWDIT-2026-5503	Project Name:	Implement Client Script and UI Policy

Technical Architecture Diagram



Architecture Guidelines Checklist:

- Includes all client and server process logic modules (Client Scripting & UI Policy triggers).
- Infrastructural demarcation clearly defined (Browser/Client vs. ServiceNow Cloud Platform).
- Data storage components mapping relational incident tables and sys_attachment engine.

Table-1: Component & Technology Stack Details

S.No	Component	Description	Technology
1.	User Interface	Interactive portal for IT agents to view, edit, and manage incidents dynamically.	ServiceNow Service Portal, HTML5, CSS3, AngularJS
2.	Application Logic-1 (Client Script)	Client-side automation executing `onLoad`, `onChange`, `onSubmit`, and `onCellEdit` handlers.	JavaScript (ES6), g_form API, g_user API
3.	Application Logic-2 (UI Policy)	Declarative rules mandating fields, setting visibility, and disabling inputs based on conditions.	ServiceNow UI Policy Engine & Policy Actions
4.	Application Logic-3 (Server Script)	Server-side verification enforcing business logic, database writes, and security policies.	JavaScript, GlideRecord, Business Rules, Script Includes

S.No	Component	Description	Technology
5.	Database Layer	Relational database storing incidents, audit trails, user profiles, and assignment groups.	ServiceNow Relational DB Engine (MariaDB underlying)
6.	Cloud Database / Storage	Cloud-hosted multi-tenant SaaS storage managed by ServiceNow High-Availability data centers.	ServiceNow Cloud Storage Engine
7.	File Storage	Storage for attachments (screenshots, logs, documentation) linked to incident records.	ServiceNow `sys_attachment` DB Table
8.	External API-1	Inbound REST endpoints to receive automated alerts from external monitoring software.	REST API, JSON, OAuth 2.0
9.	Machine Learning Model	Predictive Intelligence auto-categorizing incident urgency based on description text.	ServiceNow Predictive Intelligence (ML Engine)
10.	Infrastructure (Server/Cloud)	Deployment on ServiceNow Personal Developer Instance (PDI) / Enterprise Cloud Sub-instance.	ServiceNow Enterprise Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology / Implementation
1.	Open-Source & Standard Frameworks	Leverages standard web standards, JavaScript runtime libraries, and AngularJS framework components.	JavaScript (ECMAScript 2021), AngularJS, Bootstrap CSS
2.	Security Implementations	Role-Based Access Control (RBAC), Access Control Lists (ACLs), TLS encryption in transit, and server-side script validation.	ServiceNow ACL Engine, HTTPS/TLS 1.3, SHA-256, IAM Roles
3.	Scalable Architecture	Multi-tier cloud architecture isolating presentation (UI Policies), logic (Client/Server Scripts), and persistence layers.	ServiceNow Multi-Instance Cloud Architecture
4.	Availability	High-availability redundant architecture with automated failover and database mirroring across cloud regions.	ServiceNow Advanced High Availability (AHA) Infrastructure
5.	Performance	Minimized client-server roundtrips by handling UI validation locally via UI Policies instead of heavy server scripts.	Client-side DOM Caching, Async GlideAjax, CDN Edge distribution

References

- <https://developer.servicenow.com/> (Client Scripting & UI Policy Documentation)
- <https://c4model.com/> (Software Architecture Diagrams)
- <https://www.ibm.com/cloud/architecture> (Cloud Architecture Patterns)